

Real-Time TDDFT in RMG-DFT: Theory, Algorithm, and Implementation

*Finite vs. Periodic Systems, Velocity Gauge, Pseudopotential Corrections,
and Gaps between Theory and Code*

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Abstract

This document provides a self-contained reference for the real-time time-dependent density functional theory (RT-TDDFT) implementation in RMG-DFT. It covers the theoretical foundations at graduate-student level—with special attention to readers from the quantum-chemistry community (Gaussian basis sets, effective core potentials)—the numerical propagation algorithm, and the actual C++ implementation. A recurring theme is the *extension from finite to periodic systems*: the position operator $\hat{r}$ is ill-defined under periodic boundary conditions, forcing use of the *velocity gauge* and the vector potential $\mathbf{A}(t)$ for crystalline solids. Norm-conserving pseudopotentials (NCP) modify the velocity operator via the commutator $i[V_{\text{NL}}, \hat{r}]$; we derive this correction in full detail. The document concludes with a critical analysis of eight known gaps between the theoretical formalism and the current RMG code, together with directions for future development (Ehrenfest dynamics, non-collinear spin RT-TDDFT).

1 Introduction

Real-time TDDFT propagates the time-dependent Kohn-Sham (TDKS) equations directly in the time domain, giving access to electronic spectra, non-linear optical responses, ultrafast electron dynamics driven by intense laser fields, and (with the addition of nuclear forces) non-adiabatic dynamics. The method was formulated for molecular systems in Gaussian-basis codes (NWChem, Gaussian) and extended to periodic solids via real-space and plane-wave implementations (Octopus [1], Yabana-Bertsch [2]).

RMG-DFT is a production-grade DFT code that solves the Kohn-Sham equations on a real-space grid using multigrid acceleration, supporting CPU and GPU (CUDA, HIP, SYCL) execution [3, 4]. This document focuses on the RT-TDDFT extension resident in TDDFT/RMG_TDDFT/ and TDDFT/ELDYN/.

The structure of this document is:

- Sec. 2: theoretical foundations—TDKS, density matrix, gauge choices, current density, Bloch k-point corrections.
- Sec. 3: pseudopotential (PP) theory in depth—from the chemistry perspective, the Kleinman-Bylander separable form, and the full derivation of how PP modifies the current operator.
- Sec. 4: the predictor-corrector Magnus propagation algorithm.
- Sec. 5: RMG-specific implementation details— buffer layout, distributed matrix parallelism, observable computation.
- Sec. 6: comparative table, finite vs. periodic.
- Sec. 7: eight identified gaps between theory and code.
- Sec. 8: future directions.

2 Theoretical Foundations

2.1 Time-Dependent Kohn-Sham Equations

The time-dependent Kohn-Sham (TDKS) equations read (atomic units throughout, $\hbar = m_e = e = 1$):

$$\mathrm{i} \frac{\partial \psi_n(\mathbf{r}, t)}{\partial t} = H_{\text{KS}}[\rho(t)] \psi_n(\mathbf{r}, t), \quad (1)$$

where the KS Hamiltonian H_{KS} depends on the instantaneous electron density

$$\rho(\mathbf{r}, t) = \sum_{n=1}^{N_{\text{occ}}} f_n |\psi_n(\mathbf{r}, t)|^2, \quad (2)$$

with $f_n \in [0, 2]$ the orbital occupation numbers. The KS potential

$$V_{\text{KS}}[\rho] = V_{\text{ext}} + V_{\text{H}}[\rho] + V_{\text{xc}}[\rho] \quad (3)$$

is updated at each timestep through the *adiabatic approximation*: the exchange-correlation (XC) kernel is taken as the ground-state functional evaluated at the instantaneous density, ignoring memory effects in V_{xc} .

2.2 Density Matrix Formulation

Rather than propagating N_{occ} individual orbitals, RMG-TDDFT works in the basis of the fixed ground-state KS orbitals $\{\varphi_j\}$, expanding the time-dependent orbitals as

$$\psi_n(\mathbf{r}, t) = \sum_j C_{jn}(t) \varphi_j(\mathbf{r}). \quad (4)$$

The one-body *density matrix* in this basis is

$$P_{ij}(t) = \sum_n f_n C_{in}(t) C_{jn}^*(t). \quad (5)$$

Chemistry analogy. This is exactly the *MO-basis density matrix* familiar from Gaussian-basis RT-TDDFT (NWChem, Gaussian). In AO codes one propagates $\mathrm{i} S \dot{P} = [H, P]$ with overlap matrix S . Here the basis is *orthonormal* ($\langle \varphi_i | \varphi_j \rangle = \delta_{ij}$), so $S = \mathbb{I}$ and the *von Neumann equation* simplifies to

$$\mathrm{i} \frac{dP}{dt} = [H(t), P(t)]. \quad (6)$$

The electron density is reconstructed as

$$\rho(\mathbf{r}, t) = \rho_{\text{gnd}}(\mathbf{r}) + \sum_{ij} \Delta P_{ij}(t) \varphi_i(\mathbf{r}) \varphi_j^*(\mathbf{r}), \quad (7)$$

where $\Delta P_{ij} = P_{ij}(t) - f_i \delta_{ij}$ is the change from the ground-state diagonal.

2.3 External Field Coupling: Two Gauges

The choice of gauge determines how the external electromagnetic field enters H_{KS} .

2.3.1 Length Gauge (Finite Systems)

For an isolated molecule, the electric field $\mathbf{E}(t)$ couples via the scalar potential (dipole approximation):

$$V_{\text{ext}}(\mathbf{r}, t) = -e \mathbf{E}(t) \cdot \mathbf{r} = -(E_x(t)x + E_y(t)y + E_z(t)z). \quad (8)$$

The position operator $\hat{\mathbf{r}}$ is well-defined for a localized wavefunction. In RMG (EFIELD mode) this potential is tabulated on the grid with a sawtooth correction for periodic images (US_PP/init_efield.cpp).

2.3.2 Velocity Gauge (Periodic Systems)

For a crystal with periodic boundary conditions (PBC), the position operator $\hat{\mathbf{r}}$ is *not Hermitian*: a Bloch state $e^{i\mathbf{k}\cdot\mathbf{r}}$ is not normalizable on $\mathbb{R}^3$, and $\langle \psi_{n\mathbf{k}} | r_\alpha | \psi_{m\mathbf{k}} \rangle$ is not gauge-invariant. The length-gauge coupling $\mathbf{E} \cdot \mathbf{r}$ therefore cannot be used for an extended solid.

Instead, the external field is represented through the vector potential $\mathbf{A}(t)$ in the *minimal-coupling Hamiltonian*:

$$H(t) = \frac{[\hat{\mathbf{p}} + \mathbf{A}(t)]^2}{2} + V_{\text{KS}} = H_{\text{KS}} + \mathbf{A}(t) \cdot \hat{\mathbf{p}} + \frac{A^2(t)}{2}. \quad (9)$$

In the weak-field (linear-response) limit the $A^2/2$ term is dropped:

$$\boxed{H(t) \approx H_{\text{KS}} + \mathbf{A}(t) \cdot \hat{\mathbf{p}}}. \quad (10)$$

This is the velocity gauge used in RMG (VECTOR_POT mode). The neglected A^2 term is discussed in Gap 7.

Field forms. For a monochromatic field:

$$\mathbf{A}(t) = A_0 \hat{\mathbf{e}} \cos(\omega t), \quad \mathbf{E}(t) = -\frac{\partial \mathbf{A}}{\partial t} = A_0 \omega \hat{\mathbf{e}} \sin(\omega t). \quad (11)$$

For an instantaneous broadband (delta) kick, used for linear-response optical spectra:

$$\mathbf{A}(t) = A_0 \hat{\mathbf{e}} \delta(t). \quad (12)$$

The current RT-TDDFT implementation supports *only* the delta-kick (Gap 7).

2.4 Current Density

The macroscopic current density is the key observable in the velocity gauge. It is split into paramagnetic and diamagnetic parts:

$$\mathbf{J}_{\text{para}}(\mathbf{r}, t) = -i \sum_n f_n [\psi_n^* \nabla \psi_n - \psi_n \nabla \psi_n^*], \quad (13)$$

$$\mathbf{J}_{\text{dia}}(\mathbf{r}, t) = -\rho(\mathbf{r}, t) \mathbf{A}(t), \quad (14)$$

$$\mathbf{J}(\mathbf{r}, t) = \mathbf{J}_{\text{para}} + \mathbf{J}_{\text{dia}}. \quad (15)$$

In the matrix element formulation used in RMG, the spatially averaged (cell) current along direction α is:

$$J_\alpha(t) = \text{Re}[\text{Tr}(P(t) \cdot P^\alpha)], \quad (16)$$

where P^α is the momentum matrix defined in Sec. 2.5. The optical conductivity is extracted via Fourier transform:

$$\sigma(\omega) = \frac{J(\omega)}{E(\omega)}. \quad (17)$$

2.5 Momentum Matrix Elements for Bloch States

2.5.1 Finite System (Gamma Point, Real Orbitals)

The kinetic contribution to the current matrix element in direction α is:

$$P_{ij}^\alpha = i \cdot v_{\text{cell}} \cdot \langle \varphi_i | \frac{\partial}{\partial r_\alpha} | \varphi_j \rangle, \quad (18)$$

where $v_{\text{cell}} = \Omega_{\text{cell}}/N_{\text{grid}}$ is the real-space integration weight (unit-cell volume divided by number of grid points).

2.5.2 Bloch States: k-Vector Correction

For a Bloch orbital $\psi_{n\mathbf{k}}(\mathbf{r}) = e^{i\mathbf{k} \cdot \mathbf{r}} u_{n\mathbf{k}}(\mathbf{r})$, the chain rule gives:

$$\nabla \psi_{n\mathbf{k}} = e^{i\mathbf{k} \cdot \mathbf{r}} (\nabla u_{n\mathbf{k}} + i\mathbf{k} u_{n\mathbf{k}}). \quad (19)$$

Therefore the full momentum matrix element is:

$$\langle \psi_{m\mathbf{k}} | \hat{p}_\alpha | \psi_{n\mathbf{k}} \rangle = \langle u_{m\mathbf{k}} | -i\partial_\alpha + k_\alpha | u_{n\mathbf{k}} \rangle, \quad (20)$$

and the total momentum matrix with the i prefactor is:

$$P_{ij}^\alpha = i \cdot v_{\text{cell}} \cdot \langle \varphi_i | \frac{\partial}{\partial r_\alpha} + ik_\alpha | \varphi_j \rangle. \quad (21)$$

The gradient term ∇u is computed by `ApplyGradient` (finite-difference stencil). The k -correction $+ik_\alpha \psi_{n\mathbf{k}}$ is added explicitly in `VecPmatrix.cpp` (lines 215–217):

Listing 1: k -correction in `VecPmatrix.cpp`

```
1 psi_xC[i] += I_t * kptr->kp.kvec[0] * psi_C[i]; // I_t = i
2 psi_yC[i] += I_t * kptr->kp.kvec[1] * psi_C[i];
3 psi_zC[i] += I_t * kptr->kp.kvec[2] * psi_C[i];
```

2.5.3 Total Momentum Matrix: Kinetic + Nonlocal PP

Because the Hamiltonian contains a nonlocal pseudopotential V_{NL} , the velocity operator is *not* simply $\hat{p}$. The full derivation is given in Sec. 3.5. The result is:

$$P_{ij}^{\alpha}(\text{total}) = P_{ij}^{\alpha}(\text{KE}) + P_{ij}^{\alpha}(V_{\text{NL}}), \quad (22)$$

where $P^{\alpha}(\text{KE})$ is computed by `VecPHmatrix` and $P^{\alpha}(V_{\text{NL}})$ is added by `CurrentNlpp`.

3 Pseudopotentials: Theory and Contribution to the Current Operator

This section provides a graduate-level introduction to norm-conserving pseudopotentials (NCPP) with emphasis on concepts familiar to quantum chemists, and derives in full detail why and how the nonlocal PP modifies the current operator.

3.1 Motivation and Chemistry Analogy

Why pseudopotentials? Core electrons (e.g., the $1s^2$ of carbon) are chemically inert—they do not participate in bonding—but they are tightly bound to the nucleus, produce rapidly oscillating wavefunctions that require very fine grids, and are expensive to represent. A pseudopotential (PP) replaces the nucleus + core-electron system by a smooth effective potential that acts only on the *valence* electrons.

Analogy with Effective Core Potentials (ECPs). Quantum chemists are familiar with ECPs (Stuttgart, LANL2DZ). The conceptual idea is identical: replace the core with an effective operator. The key differences are:

- (i) **Fitting target.** ECPs in Gaussian codes are fitted to reproduce valence *energy levels and atomic spectra*. PPs in real-space/plane-wave codes additionally require the pseudo-wavefunction $\varphi_{nl}^{\text{PS}}(r)$ to match the all-electron (AE) wavefunction $\varphi_{nl}^{\text{AE}}(r)$ for $r > r_c$ (the *core radius*).
- (ii) **Norm-conservation constraint** (see below).
- (iii) **Form.** ECPs are usually fitted as Gaussian expansions times angular projectors. PPs in grid codes are most naturally expressed on a radial grid.

3.2 Norm-Conserving Pseudopotentials (NCPP)

An NCPP [5, 6] is constructed to satisfy four conditions for each angular-momentum channel (n, l) :

- (1) Same valence eigenvalues: $\epsilon_{nl}^{\text{PS}} = \epsilon_{nl}^{\text{AE}}$.
- (2) Agreement outside core radius: $\varphi_{nl}^{\text{PS}}(r) = \varphi_{nl}^{\text{AE}}(r)$ for $r > r_c$.

(3) Norm conservation inside core:

$$\int_0^{r_c} |\varphi_{nl}^{\text{PS}}(r)|^2 r^2 dr = \int_0^{r_c} |\varphi_{nl}^{\text{AE}}(r)|^2 r^2 dr. \quad (23)$$

(4) Logarithmic derivative continuity at r_c (energy transferability).

Condition (3)—norm conservation—guarantees that the pseudo-charge density equals the AE charge density everywhere, which ensures the correct linear response and scattering properties [5].

Comparison with USPP. Ultrasoft pseudopotentials (USPP) [7] relax condition (3), allowing φ^{PS} to be much smoother (hence “ultrasoft”) at the cost of introducing augmentation charges Q_{ij} to restore charge conservation. This modifies the overlap operator to $S = \mathbb{K} + \sum_{ij} Q_{ij} |\beta_i\rangle\langle\beta_j|$ and the von Neumann equation becomes $iS\dot{P} = [H, P]$ —analogous to the AO-basis Liouville equation with a non-unit overlap. The current operator also acquires additional augmentation terms (see Sec. 3.6). **RMG uses NCPP only**; all expressions below are exact for NCPP.

3.3 Semilocal Form and the Kleinman-Bylander Transformation

The raw PP, as constructed, is *semilocal*: it has a different radial dependence for each angular momentum l , but is local within each l channel:

$$V_{\text{PP}}(\mathbf{r}, \mathbf{r}') = V_{\text{local}}(r) \delta(\mathbf{r} - \mathbf{r}') + \sum_l \hat{P}_l \delta V_l(r) \delta(\mathbf{r} - \mathbf{r}'), \quad (24)$$

where $\hat{P}_l = \sum_m |Y_{lm}\rangle\langle Y_{lm}|$ is the projector onto angular momentum l and $\delta V_l = V_l - V_{\text{local}}$.

Cost problem. Computing $\langle\varphi_i|V_{\text{PP}}|\varphi_j\rangle$ for all (i, j) pairs from the semilocal form requires $O(N_{\text{states}}^2 \times N_{\text{grid}})$ operations—prohibitively expensive.

Kleinman-Bylander (KB) transformation. Kleinman and Bylander [8] showed that the semilocal operator can be rewritten as a *fully nonlocal separable* (rank-1 per channel) form:

$$V_{\text{NL}} = \sum_I \sum_{lm} D_{lm}^I |\beta_{lm}^I\rangle\langle\beta_{lm}^I|, \quad (25)$$

where the index I runs over atoms, (l, m) over angular-momentum channels, and the KB projectors and coefficients are:

$$|\beta_{lm}^I\rangle = \delta V_l^I(r) \varphi_{nl}^{\text{PS}, I}(\mathbf{r} - \mathbf{R}_I) Y_{lm}(\hat{r}_I), \quad (26)$$

$$D_{lm}^I = \frac{1}{\langle\varphi_{nl}^{\text{PS}, I}|\delta V_l^I|\varphi_{nl}^{\text{PS}, I}\rangle}. \quad (27)$$

The cost drops to $O(N_{\text{states}} \times N_{\text{proj}} \times N_{\text{grid}})$: compute $\langle\beta_{lm}|\varphi_j\rangle$ once per projector per orbital, then assemble.

RMG storage. RMG stores the projected overlaps $\langle \beta_{lm} | \varphi_j \rangle$ in `kpnr->newsint_local` and applies them via `AppNls / AppNls_0xyz`.

3.4 Full Hamiltonian with NCPP

The complete KS Hamiltonian in the NCPP/KB approximation is:

$$H = \underbrace{-\frac{1}{2}\nabla^2}_T + \underbrace{V_{\text{local}}(r) + V_{\text{H}}[\rho] + V_{\text{xc}}[\rho]}_{V_{\text{loc}} \text{ (scalar, on grid)}} + \underbrace{\sum_{I,lm} D_{lm}^I |\beta_{lm}^I\rangle \langle \beta_{lm}^I|}_{V_{\text{NL}}}. \quad (28)$$

The matrix element in the KS basis is:

$$H_{ij} = T_{ij} + \langle \varphi_i | V_{\text{loc}} | \varphi_j \rangle + \underbrace{\sum_{I,lm} D_{lm}^I \langle \varphi_i | \beta_{lm}^I \rangle \langle \beta_{lm}^I | \varphi_j \rangle}_{\text{computed by AppNls}}. \quad (29)$$

Chemistry parallel. This is the real-space analogue of $\langle \chi_i | \text{ECP} | \chi_j \rangle$ in a Gaussian basis, but evaluated by numerical quadrature on the real-space grid rather than analytically.

3.5 Why V_{NL} Modifies the Current Operator

3.5.1 Heisenberg Equation of Motion

The physical current density operator is the *velocity* operator, defined through the Heisenberg equation of motion for the position operator:

$$\hat{v}_\alpha = \frac{d\hat{r}_\alpha}{dt} = i[H, \hat{r}_\alpha], \quad \alpha \in \{x, y, z\}. \quad (30)$$

Splitting $H = H_{\text{local}} + V_{\text{NL}}$:

Local part. Since V_{loc} is a multiplicative operator, it commutes with $\hat{r}_\alpha$: $[V_{\text{loc}}, \hat{r}_\alpha] = 0$. The kinetic term gives the familiar result:

$$i[T, \hat{r}_\alpha] = -i\partial_\alpha = \hat{p}_\alpha. \quad (31)$$

Nonlocal part—key commutator. The separable V_{NL} is *not* multiplicative, so it does not commute with $\hat{r}_\alpha$:

$$i[V_{\text{NL}}, \hat{r}_\alpha] = i \sum_{I,lm} D_{lm}^I [|\beta_{lm}^I\rangle \langle \beta_{lm}^I|, \hat{r}_\alpha]. \quad (32)$$

Using the operator identity $[AB, C] = A[B, C] + [A, C]B$:

$$\begin{aligned} [|\beta\rangle \langle \beta|, \hat{r}_\alpha] &= |\beta\rangle \langle \beta| \hat{r}_\alpha - \hat{r}_\alpha |\beta\rangle \langle \beta| \\ &= |\beta\rangle \langle r_\alpha \beta| - |r_\alpha \beta\rangle \langle \beta|, \end{aligned} \quad (33)$$

where we defined $|r_\alpha\beta\rangle \equiv \hat{r}_\alpha|\beta\rangle$, i.e., the projector multiplied pointwise by the coordinate r_α .

The *full velocity operator* is therefore:

$$\hat{v}_\alpha = \hat{p}_\alpha + i \sum_{I,lm} D_{lm}^I \left(|\beta_{lm}^I\rangle \langle r_\alpha \beta_{lm}^I| - |r_\alpha \beta_{lm}^I\rangle \langle \beta_{lm}^I| \right). \quad (34)$$

3.5.2 Matrix Element of the NL-PP Correction

Taking the matrix element between KS orbitals φ_i and φ_j :

$$\langle \varphi_i | i [V_{\text{NL}}, \hat{r}_\alpha] | \varphi_j \rangle = i \sum_{I,lm} D_{lm}^I \left[\underbrace{\langle \varphi_i | \beta_{lm}^I \rangle}_{\text{overlap}} \underbrace{\langle r_\alpha \beta_{lm}^I | \varphi_j \rangle}_{\text{coord-weighted proj.}} - \underbrace{\langle \varphi_i | r_\alpha \beta_{lm}^I \rangle}_{\text{coord-weighted proj.}} \underbrace{\langle \beta_{lm}^I | \varphi_j \rangle}_{\text{overlap}} \right]. \quad (35)$$

How RMG computes Eq. (35). The key observation is that $\langle r_\alpha \beta | \varphi \rangle = \langle \varphi | r_\alpha \beta \rangle^*$ for real β and complex φ . Both terms therefore involve the quantity $\langle \varphi_i | r_\alpha \delta V_l \varphi^{\text{PS}} | \dots \rangle$.

`AppNls_Oxyz` with direction flag `ixyz=1,2,3` computes, for a block of orbitals φ_j (index j from `st_start` to `st_start+this_block_size`):

$$\text{nv}[r] = \sum_{I,lm} D_{lm}^I \beta_{lm}^I(\mathbf{r}) r_\alpha \cdot \langle \beta_{lm}^I | \varphi_j \rangle, \quad (36)$$

i.e., the KB projector β weighted by the Cartesian coordinate r_α and multiplied by the pre-computed overlap $\langle \beta | \varphi_j \rangle$.

In `CurrentNlpp.cpp`, after calling `AppNls_Oxyz` for each direction, a GEMM computes:

$$\text{block_matrix}_\alpha = v_{\text{cell}} \underbrace{[\varphi_{\text{block}}]^\dagger}_{\text{bra}} \cdot \underbrace{\text{nv}}_{\text{ket}}, \quad (37)$$

which gives $v_{\text{cell}} \cdot \langle \varphi_i | r_\alpha V_{\text{NL}}^\dagger | \varphi_j \rangle$ for the block. Multiplying by i and adding to `Pxmatrix` accumulates the NL-PP correction onto the kinetic-only P^α from `VecPHmatrix`:

Listing 2: Accumulation in `CurrentNlpp.cpp` (upper triangle)

```
1 kptr->Pxmatrix_cpu[...] += I_t * block_matrix_x[...]; // I_t = i
2 kptr->Pymatrix_cpu[...] += I_t * block_matrix_y[...];
3 kptr->Pzmatrix_cpu[...] += I_t * block_matrix_z[...];
```

3.5.3 Hermiticity Requirement

The velocity operator $\hat{v}_\alpha$ must be Hermitian ($\hat{v}_\alpha^\dagger = \hat{v}_\alpha$) to guarantee that the current $J_\alpha = \text{Re}[\text{Tr}(P\hat{v}_\alpha)] \in \mathbb{R}$.

Kinetic contribution. The gradient operator is anti-Hermitian: $(-i\nabla)^\dagger = -i\nabla$, so $\hat{p}_\alpha = -i\partial_\alpha$ is Hermitian. In matrix form: $\langle \varphi_i | \partial_\alpha | \varphi_j \rangle^* = -\langle \varphi_j | \partial_\alpha | \varphi_i \rangle$ (after integration by parts), so ∇ is anti-Hermitian. Multiplying by i gives a Hermitian matrix.

NL-PP contribution. From Eq. (33): interchanging $i \leftrightarrow j$ and complex-conjugating gives exactly -1 times the original expression, confirming that $[V_{\text{NL}}, \hat{r}_\alpha]$ is anti-Hermitian. Multiplying by i produces a Hermitian contribution to $\hat{v}_\alpha$.

Symmetrization in RMG. `CurrentNlpp` computes only the *upper triangle* of P^α (block rows $\geq$ block column for the tiledMM path, or the (ib, jb) block structure for ScaLAPACK). The lower triangle is filled by `MyConj`: $P_{ji}^\alpha = (P_{ij}^\alpha)^*$, enforcing the required Hermitian symmetry:

Listing 3: Hermitian symmetrization in `CurrentNlpp.cpp` (lower triangle)

```
1 kptr->Pxmatrix_cpu[...] += MyConj(I_t * block_matrix_x[...]);
```

3.6 Comparison: NCPP vs USPP vs PAW

Although RMG uses NCPP, the reader may encounter USPP or PAW in other codes. Table 1 summarizes the key differences relevant for RT-TDDFT.

Table 1: Comparison of pseudopotential types relevant for RT-TDDFT.

Aspect	NCPP	USPP	PAW
Norm conservation	Yes	No	Implicit (AE)
Augmentation charges	No	Yes (Q_{ij})	Yes (on-site)
Overlap matrix S	$\nless$	$\neq \nless$	$\neq \nless$
von Neumann equation	$i\dot{P} = [H, P]$	$iS\dot{P} = [H, P]$	$iS\dot{P} = [H, P]$
Current correction	$i[V_{\text{NL}}, \hat{r}]$	$+Q$ -augmented	$+ \text{on-site AE}$
Chemistry analogue	ECP (approx.)	No direct analogue	No direct analogue
RMG support	Yes	No	No

USPP Liouville equation. For USPP the Liouville equation $iS\dot{P} = [H, P]$ is structurally identical to the *AO-basis equation* familiar from Gaussian codes: $iS\dot{\sigma} = [F, \sigma]$ where F is the Fock/KS matrix and σ is the AO density matrix. The key point for the quantum chemist: in both cases, $S \neq \nless$ forces a more expensive propagator and modifies the current.

3.7 Pseudopotential Contribution to Forces (Ehrenfest Preview)

For completeness—and as a preview of Ehrenfest dynamics—the NL-PP contribution to the force on nucleus I along direction α is:

$$F_{\alpha, \text{NL}}^I = -\frac{\partial E_{\text{NL}}}{\partial R_\alpha^I} = -\sum_{ij} P_{ij} \sum_{lm} D_{lm}^I \left[\left\langle \frac{\partial \varphi_i}{\partial R_\alpha^I} \middle| \beta_{lm}^I \right\rangle \left| \beta_{lm}^I \right\rangle \varphi_j + \text{h.c.} \right], \quad (38)$$

where the derivative $\partial \beta_{lm}^I / \partial R_\alpha^I$ arises because the projector $\beta_{lm}^I(\mathbf{r} - \mathbf{R}_I)$ moves with the nucleus. In RMG, the nonlocal force contribution is evaluated in `Force/CorrectForces.cpp`. Ehrenfest dynamics will require calling this force routine with the *time-dependent* density matrix $P(t)$ at each timestep.

4 Propagation Algorithm

4.1 Magnus Expansion

The formal solution to the von Neumann equation Eq. (6) is

$$P(t + \Delta t) = e^{-i\Omega} P(t) e^{+i\Omega}, \quad (39)$$

where Ω is given by the Magnus expansion [9]:

$$\Omega = \Omega_1 + \Omega_2 + \dots, \quad \Omega_1 = \int_t^{t+\Delta t} H(t') dt'. \quad (40)$$

To first order (midpoint rule), $\Omega_1 \approx \frac{1}{2}[H(t) + H(t + \Delta t)]\Delta t$. **Key property:** the Magnus propagator is exactly unitary regardless of Δt (unlike simple Euler integration), guaranteeing conservation of particle number and orthonormality.

4.2 Predictor-Corrector Scheme

Because $H(t + \Delta t)$ depends on $P(t + \Delta t)$ (via $\rho \rightarrow V_{\text{KS}} \rightarrow H$), RMG uses a self-consistent predictor-corrector loop:

Step 1. Predictor (linear extrapolation, `extrapolate.Hmatrix`):

$$H_{\text{pred}}(t + \Delta t) = 2H(t) - H(t - \Delta t). \quad (41)$$

Step 2. SCF corrector loop (convergence when $\|H_{\text{new}} - H_{\text{old}}\|_{\infty} < 10^{-7}$):

(a) **Magnus operator** (`magnus`):

$$\Omega = \frac{1}{2}[H(t) + H_{\text{pred}}(t + \Delta t)]\Delta t. \quad (42)$$

(b) **Propagate density matrix** (`eldyn.ort` $\rightarrow$ `commutp` or `diagev`):

$$P(t + \Delta t) = e^{-i\Omega} P(t) e^{+i\Omega}. \quad (43)$$

(c) **New density** (`GetNewRho_rmgtddft`): $\rho(t + \Delta t)$ from $\Delta P = P(t + \Delta t) - \text{diag}(f_i)$.

(d) **Update potentials:** V_{H} via Poisson solver, V_{xc} via adiabatic functional.

(e) **Update Hamiltonian** (`HmatrixUpdate`):

$$H(t + \Delta t) += v_{\text{cell}} \sum_j \langle \varphi_i | \Delta V | \varphi_j \rangle. \quad (44)$$

Step 3. Advance: $P(t) \leftarrow P(t + \Delta t)$; $H(t - \Delta t) \leftarrow H(t)$; $H(t) \leftarrow H(t + \Delta t)$.

4.3 BCH Propagator (`commutp`)

The Baker-Campbell-Hausdorff (BCH) identity expresses the similarity transform in Eq. (43) as a nested commutator series:

$$e^{-i\Omega} P e^{+i\Omega} = \sum_{k=0}^{\infty} \frac{1}{k!} (-i)^k \underbrace{[\Omega, [\Omega, \dots [\Omega, P] \dots]]}_{k \text{ nested commutators}}. \quad (45)$$

RMG stores P split into real symmetric (S) and antisymmetric (A) parts: $P = S + iA$, packed as a single block of size $2N^2$ doubles. Each iteration computes

$$P^{(k)} = P^{(k-1)} + \frac{1}{k} [\Omega, P^{(k-1)}], \quad (46)$$

alternating between `commatr` (real correction to A) and `commstr` (imaginary correction to S), until $\|P^{(k)} - P^{(k-1)}\|$ falls below tolerance.

4.4 Diagonalization Propagator (`diagev`) — Alternative

An alternative propagator (selected by `Ieldyn=2`) diagonalizes Ω : $\Omega = V \text{diag}(\varepsilon_k) V^\dagger$, then:

$$P(t + \Delta t) = V e^{-i \text{diag}(\varepsilon_k) \Delta t} V^\dagger P(t) V e^{+i \text{diag}(\varepsilon_k) \Delta t} V^\dagger. \quad (47)$$

This is exact but requires a real symmetric Ω (Gamma-point, real orbitals). For complex k-point Hamiltonians only the BCH propagator is used.

5 RMG Implementation

5.1 Basis, Orbital Storage, and Grid

The ground-state KS orbitals $\{\varphi_j\}$ are stored as real-space grid functions on the *coarse grid* (wavefunction grid). The fine grid (density grid) has a higher resolution by the ratio `default_FG_RATIO` in each direction.

The real-space integration weight is:

$$v_{\text{cell}} = \frac{\Omega_{\text{cell}}}{N_x N_y N_z}, \quad (48)$$

where N_x, N_y, N_z are the number of coarse-grid points in each direction. All matrix-element GEMMs carry this prefactor.

Template types encode the physics:

OrbitalType	MatrixType	Scenario
double	double	Γ -point + EFIELD
double	complex<double>	Γ -point + VECTOR_POT
complex<double>	complex<double>	k-points + VECTOR_POT

5.2 EFIELD Mode (Finite Systems)

A delta-kick in the length gauge applies:

$$V_{\text{kick}}(\mathbf{r}) = -\mathbf{E}_0 \cdot \mathbf{r}, \quad H(t=0) += \frac{1}{\Delta t} \langle \varphi_i | V_{\text{kick}} | \varphi_j \rangle, \quad (49)$$

where the factor $1/\Delta t$ converts the instantaneous kick into a finite-amplitude perturbation. The dipole moment $\mathbf{d}(t) = \int \mathbf{r} \rho(\mathbf{r}, t) d\mathbf{r}$ is the observable.

5.3 VECTOR_POT Mode (Periodic Systems)

Pre-computation (once, at $t = 0$).

1. `VecPHmatrix`: compute $P_{ij}^\alpha(\text{KE})$ [Eq. (21)] via finite-difference gradient + k-correction.
2. `CurrentNlpp`: add $P_{ij}^\alpha(V_{\text{NL}})$ [Eq. (35)] via `AppNls_0xyz`.
3. **Result**: `kptr->Pxmatrix_cpu`, `Pymatrix_cpu`, `Pzmatrix_cpu` contain the total P^α [Eq. (22)].

Initial kick at $t = 0$.

$$H(t=0) += \sum_{\alpha} A_0^{\alpha} P^{\alpha}, \quad (50)$$

applied to both `Hmatrix_0_cpu` and `Hmatrix_m1_cpu` (guard: `pre_steps == 0`).

Current observable. At each timestep, for each k-point:

$$J_{\alpha}(t) = w_k \text{Re}[\text{Tr}(P(t) \cdot P^{\alpha})] = w_k \text{Re}[\text{zdotc}(\text{Pn0}, \text{Pxmatrix})]. \quad (51)$$

Brillouin zone average: $J_{\alpha} = \sum_k J_{\alpha}^{(k)}$ with MPI reduction over `kpsub_comm` and `eldyn_comm`.

5.4 Buffer Layout and Aliasing

Table 2: Key density-matrix and Hamiltonian buffers in RMG RT-TDDFT. `Hmatrix_m1_cpu` serves three sequential roles within a single timestep.

Buffer	Size	Role
<code>Pn0_cpu</code>	$2N^2$	$P(t)$ (current density matrix)
<code>Pn1_cpu</code>	$2N^2$	$P(t + \Delta t)$ (updated each SCF iter)
<code>Hmatrix_0_cpu</code>	N^2	$H(t)$
<code>Hmatrix_m1_cpu</code>	N^2	$H(t - \Delta t)$ then Ω then $\langle \Delta V \rangle$
<code>Hmatrix_1_cpu</code>	N^2	$H_{\text{pred}}(t + \Delta t)$
<code>Hmatrix_cpu</code>	N^2	Accumulator for $H(t + \Delta t)$

The triple aliasing of `Hmatrix_m1_cpu` is intentional but fragile (Gap 7).

Table 3: Comparison of the EFIELD (finite) and VECTOR_POT (periodic) RT-TDDFT modes in RMG.

Aspect	EFIELD (finite)	VECTOR_POT (periodic)
OrbitalType	double	double/complex<double>
MatrixType	double	complex<double>
GEMM adjoint	"t" (transpose)	"c" (conj. transpose)
Gauge	Length: $V = -\mathbf{E} \cdot \mathbf{r}$	Velocity: $H = H_{\text{KS}} + \mathbf{A} \cdot \hat{\mathbf{p}}$
Perturbation	V_{kick} on grid	$A_0 P^\alpha$ matrix
Observable	Dipole $\mathbf{d}(t)$	Current $J_\alpha(t)$
k-point sum	N/A (Gamma only)	$\sum_k w_k J_\alpha^{(k)}$
NL-PP correction	Not needed for dipole	Required for J
Position operator $\hat{r}$	Well-defined	Ill-defined under PBC
Output file	*_dipole.dat	*_current.dat

6 Finite vs. Periodic: Comparative Summary

7 Known Gaps: Theory vs. Implementation

The following gaps were identified by tracing the C++ source against the theoretical formalism. Each gap is labelled with a severity: **Critical** (gives wrong physics in named scenarios), **Moderate** (limitation or fragility), **Minor** (cosmetic/documentation).

[GAP-1] CRITICAL — VECTOR_POT implements delta-kick only

Theory: The velocity-gauge Hamiltonian should evolve as $H(t) = H_{\text{KS}} + \mathbf{A}(t) \cdot \hat{\mathbf{p}}$ with a general $\mathbf{A}(t)$, e.g., $\mathbf{A}(t) = A_0 \cos(\omega t) \hat{\mathbf{e}}$ for a CW laser.

Code: The block that would add $\mathbf{A}(t) \cos(\omega t) \cdot P^\alpha$ to H at each timestep is commented out (RmgTddft.cpp, lines 694–701), and `ct.tddft_frequency` is never used in the propagation loop.

Consequence: Linear-response optical spectra (FT of $J(t)$ after a delta-kick) are valid. CW, pulse, or non-perturbative field simulations are not possible.

Fix required: Uncomment and correct the `daxpy` block that adds $A_0 \cos(\omega t_{\text{step}}) P^\alpha$ to `Hmatrix_0_cpu` at each time step.

[GAP-2] MODERATE — Diamagnetic current absent

Theory: $J_{\text{total}} = J_{\text{para}} + J_{\text{dia}}$, with $J_{\text{dia}} = -\rho(\mathbf{r}, t) \mathbf{A}(t)$ needed for gauge invariance under a sustained field.

Code: Only $J_{\text{para}} = \text{Re}[\text{Tr}(P \cdot P^\alpha)]$ is computed [Eq. (51)].

Current status: Harmless for the delta-kick ($\mathbf{A}(t) = 0$ for $t > 0$). Must be added before GAP-1 can be fixed.

[GAP-3] MODERATE — A^2 term dropped, undocumented

Theory: The full minimal-coupling Hamiltonian [Eq. (9)] contains $A^2(t)/2$.

Code: Only the linear term $\mathbf{A} \cdot \hat{\mathbf{p}}$ is included. No amplitude check enforces the weak-field ($A \ll 1$) regime.

Consequence: The diamagnetic energy shift $A^2\rho/2$ is missing; results are physically correct only in the perturbative (linear-response) regime.

[GAP-4] CRITICAL — Non-collinear spin incomplete in VECTOR_POT path

Theory: For non-collinear spin, V_{xc} is a 2×2 matrix in spin space with components (c_x, c_y, c_z) (off-diagonal spin terms from the Zeeman-like XC field).

Code: `HmatrixUpdate` uses only the scalar `vtot` (spin-up or spin-down diagonal) in the VECTOR_POT path. Two comments labelled “noncoll need change” mark the exact locations in the code.

Consequence: Spin dynamics with the vector potential give a silently wrong $H(t)$ — the XC spin-orbit-like off-diagonal terms are absent.

Fix: Extend `HmatrixUpdate` to construct the full 2×2 `vtot` spinor and apply it in the GEMM, analogous to the scalar path.

[GAP-5] MODERATE — `Hmatrix_m1_cpu` triple aliasing

Within a single timestep, `Hmatrix_m1_cpu` serves three sequential roles: (1) $H(t - \Delta t)$ read by `extrapolate_Hmatrix`; (2) Ω written by `magnus`; (3) $\langle \Delta V \rangle$ written by `HmatrixUpdate`. This is not a current bug—the sequential order is correct—but it is a hazard for any future developer who reorders or parallelizes these calls. See Table 2.

[GAP-6] MINOR — `Hmatrix_cpu` implicit accumulation

`Hmatrix_cpu` is never explicitly reset to $H(t)$ at the start of each timestep. Correctness relies on `vtot` always containing the *incremental* potential change between SCF iterations. If this convention is ever broken (e.g., by resetting `vtot` to the full potential), the accumulated H will be incorrect.

[GAP-7] MODERATE — No mode/boundary-condition guard

No assertion or error check prevents the user from selecting EFIELD with k-points or VECTOR_POT with `is_gamma = true`. Both combinations produce unphysical results silently. A guard at startup (e.g., in `RmgTddft.cpp`) should enforce the correct pairing.

[GAP-8] MINOR — `efield_tddft_crds` dual meaning

In EFIELD mode, `ct.efield_tddft_crds[0,1,2]` is the electric field amplitude $E_0\hat{\mathbf{e}}$. In VECTOR_POT mode it is the vector potential amplitude $A_0\hat{\mathbf{e}}$. The physical relationship $E_0 = A_0\omega$ is nowhere documented in the input handler, so a user who sets the same

numerical value in both modes applies perturbations that differ by a factor ω .

8 Future Directions

8.1 Sustained Field / CW Laser (fixing GAP-7)

The most important near-term fix is to complete the velocity-gauge time loop by uncommenting and correcting the $\mathbf{A}(t) \cos(\omega t) \cdot P^\alpha$ update. This enables:

- Resonant excitation and Rabi oscillations.
- Non-linear (high-harmonic) response.
- Time-domain pump-probe simulations.

Adding the diamagnetic current (GAP-2) is a prerequisite.

8.2 Non-Collinear Spin RT-TDDFT (fixing GAP-4)

Extending `HmatrixUpdate` to the full 2×2 spin-space Hamiltonian enables spin dynamics—the primary scientific goal of the current development effort. The pseudo-spinor orbitals $\Psi_n = (\psi_{n\uparrow}, \psi_{n\downarrow})^T$ carry two-component wavefunctions on the grid; the density matrix becomes a $2N \times 2N$ block matrix and P^α must be extended accordingly.

8.3 Ehrenfest Dynamics

Ehrenfest dynamics couples the electronic $P(t)$ propagation to classical nuclear motion:

$$M_I \ddot{\mathbf{R}}_I = -\nabla_{\mathbf{R}_I} E[\rho(t)], \quad (52)$$

where the force includes the time-dependent NL-PP term of Eq. (38). The key implementation challenge is updating the fixed KS orbital basis $\{\varphi_j\}$ as nuclei move: the projectors β_{lm}^I shift with $\mathbf{R}_I$, requiring a re-projection at each step or an on-the-fly Pulay correction to the force.

8.4 Spin + Ehrenfest

The combination of non-collinear spin dynamics and Ehrenfest nuclear motion constitutes the full spin-TDDFT + MD target. The P^α matrices must be extended to 2×2 spinor blocks, and the force expression must include the spin-dependent XC contribution $\partial E_{\text{xc}}^{\text{noncoll}} / \partial \mathbf{R}_I$.

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